

ONE-WAY ANOVA

Q1) Define the one-way ANOVA model

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{iid}{\sim} (0, \sigma^2) \quad (i=1, \dots, k) \quad (j=1, \dots, n_i)$$

μ : overall mean without the treatment effect

τ_i : Contribution of the i th treatment to the mean

Q2) What does "one-way" mean in the one-way ANOVA model?

One-way refers to the fact that the number of treatment type is one.

Q3) What would a "two-way" ANOVA model mean?

It refers to the fact that the model considers two treatment types

Q6) Prove Theorem 3

Consider the one-way ANOVA model with constraint $\tau_k = 0$

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2) \quad (i=1, 2, \dots, k)$$

In this case, the least squares estimators for $\mu, \tau_1, \dots, \tau_{k-1}$ are given by:

$$\hat{\mu} = \bar{y}_k \quad \text{and} \quad \hat{\tau}_i = \bar{y}_i - \bar{y}_k \quad (i=1, \dots, k-1)$$

$$y = \begin{bmatrix} y_{11} \\ \vdots \\ y_{1n_1} \\ \vdots \\ y_{k1} \\ \vdots \\ y_{kn_k} \end{bmatrix}, \quad X = \begin{bmatrix} 1 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 1 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 1 & 0 & \dots & 0 \end{bmatrix} = \begin{bmatrix} 1_{n_1} & 1_{n_1} & 0 & \dots & 0 \\ 1_{n_2} & 0 & 1_{n_2} & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1_{n_{k-1}} & 0 & 0 & \dots & 1_{n_{k-1}} \\ 1_{n_k} & 0 & 0 & \dots & 0 \end{bmatrix}$$

$$\beta = \begin{bmatrix} \mu \\ \tau_1 \\ \vdots \\ \tau_{k-1} \end{bmatrix} \quad \text{and} \quad \varepsilon = \begin{bmatrix} \varepsilon_{11} \\ \vdots \\ \varepsilon_{1n_1} \\ \vdots \\ \varepsilon_{k1} \\ \vdots \\ \varepsilon_{kn_k} \end{bmatrix}$$

Consider the following one-way ANOVA model:

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2), \quad i=1, 2, \quad j=1, 2, 3$$

$$y_{11} = \mu + \tau_1 + \varepsilon_{11} \quad y_{12} = \mu + \tau_1 + \varepsilon_{12} \quad y_{13} = \mu + \tau_1 + \varepsilon_{13}$$

$$y_{21} = \mu + \tau_2 + \varepsilon_{21} \quad y_{22} = \mu + \tau_2 + \varepsilon_{22} \quad y_{23} = \mu + \tau_2 + \varepsilon_{23}$$

$$y = \begin{bmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{21} \\ y_{22} \\ y_{23} \end{bmatrix}, \quad X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \quad \beta = \begin{bmatrix} \mu \\ \tau_1 \\ \tau_2 \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \varepsilon_{23} \end{bmatrix}$$

Q4) In the one-way ANOVA model, the design matrix is NOT full rank.

What kind of problem would arise if we were to fit a regression model.

The matrix $X^T X$ is singular and thus non-invertible.

If the design matrix X is not full-rank, the ANOVA model is not identifiable

Q5) What constraint is imposed so that the design matrix X is full rank?

Last column of the design matrix is eliminated ($\tau_k = 0$)

$$X^T X = \begin{bmatrix} n & n_1 & n_2 & \dots & n_{k-1} \\ n_1 & n_1 & 0 & \dots & 0 \\ n_2 & 0 & n_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n_{k-1} & 0 & 0 & \dots & n_{k-1} \end{bmatrix}, \quad X^T y = \begin{bmatrix} y_{\cdot} \\ y_{1\cdot} \\ y_{2\cdot} \\ \vdots \\ y_{k-1\cdot} \end{bmatrix}, \quad y_{\cdot} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} y_{ij}, \quad y_{i\cdot} = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}$$

$$(X^T X)^{-1} \hat{\beta} = \begin{bmatrix} n & n_1 & n_2 & \dots & n_{k-1} \\ n_1 & n_1 & 0 & \dots & 0 \\ n_2 & 0 & n_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n_{k-1} & 0 & 0 & \dots & n_{k-1} \end{bmatrix}^{-1} \begin{bmatrix} \mu \\ \tau_1 \\ \vdots \\ \tau_{k-1} \end{bmatrix} = \begin{bmatrix} n\mu + n_1\tau_1 + \dots + n_{k-1}\tau_{k-1} \\ n_1\mu + n_1\tau_1 \\ n_2\mu + n_2\tau_2 \\ \vdots \\ n_{k-1}\mu + n_{k-1}\tau_{k-1} \end{bmatrix}$$

$$(n\mu + n_1\tau_1 + \dots + n_{k-1}\tau_{k-1}) - (n_1\mu + n_1\tau_1) - \dots - (n_{k-1}\mu + n_{k-1}\tau_{k-1})$$

$$= (n - n_1 - \dots - n_{k-1})\mu$$

$$= y_{\cdot} - (y_{1\cdot} + \dots + y_{k-1\cdot}) = y_k$$

$$\therefore n_k \mu = y_k \quad \therefore \hat{\mu} = \bar{y}_k$$

$$n_i \hat{\mu} + n_i \hat{\tau}_i = y_{i\cdot} \quad \therefore \hat{\tau}_i = \bar{y}_i - \bar{y}_k \quad (i=1, \dots, k-1)$$

Q7) Consider the one-way ANOVA model with constraint $\tau_k = 0$ and normality assumption. We wish to test whether there is treatment effect or not.

$$H_0: \tau_1 = \dots = \tau_{k-1} = 0 \quad \text{vs} \quad H_1: \text{not } H_0$$

Consider the test statistic F defined as below:

$$F = \frac{SSB/(K-1)}{SSW/(n-K)}, \quad SSB = \sum_{i=1}^K n_i (\bar{y}_i - \bar{y})^2 \quad \text{and} \quad SSW = \sum_{i=1}^K \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$$

Show that under H_0 : $F \sim F(K-1, n-K)$

$$SSB = \sum_{i=1}^K n_i (\bar{y}_i - \bar{y})^2 = \|(\Pi_P - \Pi_{\mathbf{1}})y\|^2 = y^T (\Pi_P - \Pi_{\mathbf{1}}) y, \quad \Pi_P = X_P (X_P^T X_P)^{-1} X_P^T$$

$$X_P = \begin{bmatrix} \mathbf{1}_{n_1} & 0 & \dots & 0 \\ 0 & \mathbf{1}_{n_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \mathbf{1}_{n_k} \end{bmatrix}$$

$$SSB/\sigma^2 = y^T \left(\frac{\Pi_P - \Pi_{\mathbf{1}}}{\sigma^2} \right) y = y^T A y$$

$$A \Sigma = \left(\frac{\Pi_P - \Pi_{\mathbf{1}}}{\sigma^2} \right) \sigma^2 I_n = \Pi_P - \Pi_{\mathbf{1}} : \text{idempotent}$$

$$r = \text{rank}(A) = \text{trace}(\Pi_P) - \text{trace}(\Pi_{\mathbf{1}}) = K-1$$

$$\delta = E[y]^T A E[y] = 0 \quad \because E[y] = 0 \quad \text{under } H_0$$

$$\therefore SSB/\sigma^2 \sim \chi^2(K-1)$$

$$SSW/\sigma^2 = y^T \left(\frac{I_n - \Pi_P}{\sigma^2} \right) y = y^T B y$$

$$B \Sigma = \left(\frac{I_n - \Pi_P}{\sigma^2} \right) \sigma^2 I_n = I_n - \Pi_P : \text{idempotent}$$

$$r = \text{rank}(B) = \text{trace}(I_n) - \text{trace}(\Pi_P) = n-K$$

$$\delta = E[y]^T B E[y] = 0 \quad \because E[y] = 0 \quad \text{under } H_0$$

$$SSW/\sigma^2 \sim \chi^2(n-K)$$

$$SSB \perp SSW \quad \because A \Sigma B = (\Pi_P - \Pi_{\mathbf{1}}) \sigma^2 I_n (I_n - \Pi_P) \\ = \sigma^2 (\Pi_P - \Pi_P - \Pi_{\mathbf{1}} + \Pi_{\mathbf{1}}) = 0$$

$$\therefore \frac{SSB/(K-1)}{SSW/(n-K)} \sim F(K-1, n-K)$$